2026 System Acceptance Review
Rice Robotics R-OWL-vers
Houston, Texas
United States

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Rice Robotics R-OWL-vers is organized into four major groups that match subsystem ownership and integration interfaces: Hardware, Infrastructure, Perception, and Control. Further separations of these groups can be seen in Fig. 1. This structure is used to manage interfaces (mechanical mounting, electrical power/communications, and ROS 2 topic/service boundaries) and supports incremental integration from independent development to field testing. With the competition, the team has worked to develop a rover completely from scratch, which we have named SAM. All systems are new and developed during the URC 2026 cycle.

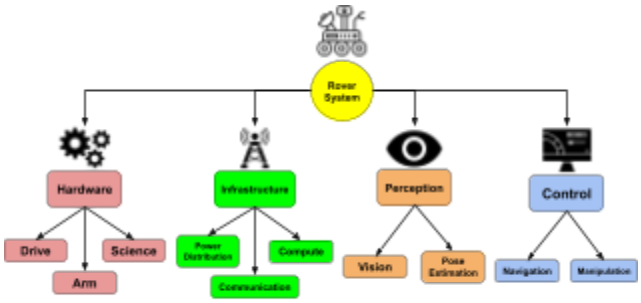

Figure 1: System Architecture

Hardware

The rover drive system utilizes a rocker-bogie suspension system composed of aluminum extrusion and revolute joints that enables dynamic navigation of large obstacles and difficult terrain. The rover body consists of sheet metal walls connected to an aluminum extrusion frame. Rubber wheels are driven by DC motors and modified with spikes to increase traction.

The rover arm system consists of a two degree-of-freedom manipulator with two revolute joints mounted to the rover on a linear slide. A two degree-of-freedom gripper is utilized to interact with the mock-up lander during equipment servicing and lift objects up to 5 kg. The arm's revolute joints utilize RobStride 02 and 04 motors, the linear slide is driven by a Nema 23 stepper motor, and the gripper is controlled via two servos. Dual parallelogram linkages are used to maintain parallelism between the base and the gripper, enabling consistent and convenient positioning of the end-effector for manipulation and equipment servicing tasks.

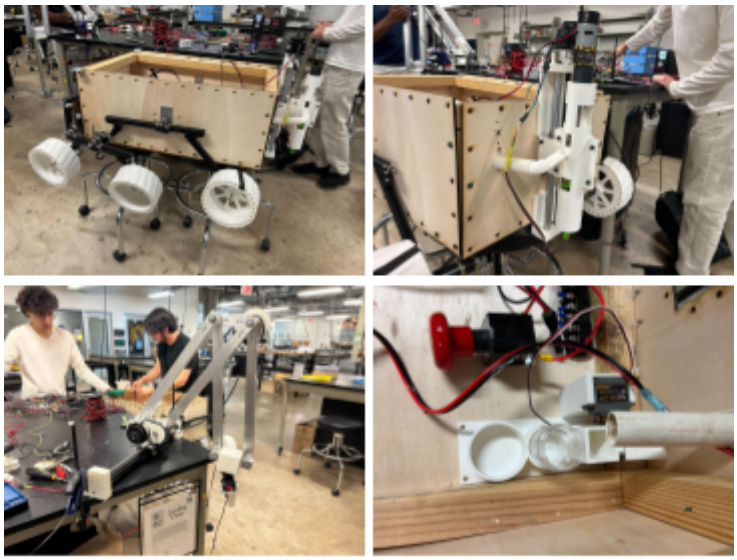

Figure 2: Mechanical Systems of the Rover.
Clockwise from top left: Body, Drill, Science Module, Arm

The rover's scientific functions are facilitated by a sample collection system and an onboard analysis module. An Archimedes screw mechanism consisting of an auger drill bit surrounded by a sleeve drills down below 10 cm; soil is carried upward along the auger threads, and the sample is ejected above the surface. This allows controlled sample collection from a specified depth. A DC motor spins the drill and a linear slide is used to move the drill vertically. Once the sample is collected within the screw mechanism, it is ejected to the science module onboard the rover where it is stored in a cache container and analyzed for evidence of life (see page 7 for detailed description of the science module).

Infrastructure

A centralized power system with a 48 V, 20 Ah LiPo battery with 40 A integrated BMS feeds a master E-stop and fused 48 V bus. DC/DC conversion provides 24 V actuation, 12 V compute/sensing,

and a 5 V logic rail for low-power peripherals. A separate 24 V, 5 Ah drive battery supplies an additional drive bus. Motor and compute ground returns are separated for noise isolation. The end goal is to have compact power distribution PCB assemblies to save on space, ensure reliability, and allow for reproducibility.

A 900 MHz RFD900x link to the onboard Jetson carries commands and telemetry (validated ~1 km NLOS; arm control over RF demonstrated), including rover health and GNSS/RTK data. A 2.4 GHz link carries video, streamed via a GStreamer and WebRTC system for real-time operator viewing.

Onboard compute centers on an NVIDIA Jetson Orin Nano running ROS 2 for perception, navigation support, and operator interfacing. Low-level actuation is handled by ESP32 controllers, with arm joints interfaced over CAN. Current operator control is keyboard-based, with plans to support a joystick-style game controller for more intuitive driving and arm teleoperation.

Figure 3: Electronics System Diagram

Perception

Sensors include three Arducam USB cameras (IMX291, 1080p, ~160° fisheye), one Intel RealSense camera (D455), and a Livox Mid-360 3D LiDAR. This lineup supports ArUco detection and pose estimation, YOLO-based object detection, LiDAR-based perception for navigation, and RealSense-based perception for arm servicing tasks.

Rover pose uses GNSS/RTK as the global reference, with planned wheel-encoder odometry and an IMU for short-term stability. Close-range navigation and servicing uses vision for relative localization. The arm uses motor encoders for joint/end-effector state, with vision-assisted alignment.

Figure 4: Software Structure Diagram

Control

Navigation is built on Nav2 and optimized for teleoperation and autonomy. Nav2 utilizes inputs from the vision and pose estimation systems to build costmaps and plan safe motion. This system outputs motion commands to the drivetrain controller and supports waypoint navigation.

Manipulation is teleop-first using ros2_control. Operator inputs are published as joint commands, translated by a small bridge layer into the controller command format, then executed by the arm velocity controller in Genesis and the real arm. MoveIt 2 remains planned for collision-aware, repeatable semi-autonomous servicing motions.

Approach to Integration

Integration follows a staged, interface-driven process consistent with our system architecture. Subsystems are first developed and verified independently (Hardware, Infrastructure, Perception, Control), then integrated incrementally at well-defined interfaces: mechanical mounting, power distribution rails, communications buses (e.g., CAN/serial/RF), and ROS 2 topics/services. Simulation through Genesis is used to validate operator workflows and control stability before field testing, then the rover is progressively transitioned from bench testing (single subsystem and infrastructure) to fully integrated system operation. We use Genesis as our physics simulation platform to validate control and

operator procedures before full physical integration. Current integration work (as of 2/27) focuses on completing infrastructure bring-up (compute and power distribution), improving operator control hardware (transitioning toward a game-controller style interface), mounting and wiring perception sensors (cameras and LiDAR), and deploying perception/control software to onboard computers for end-to-end testing.

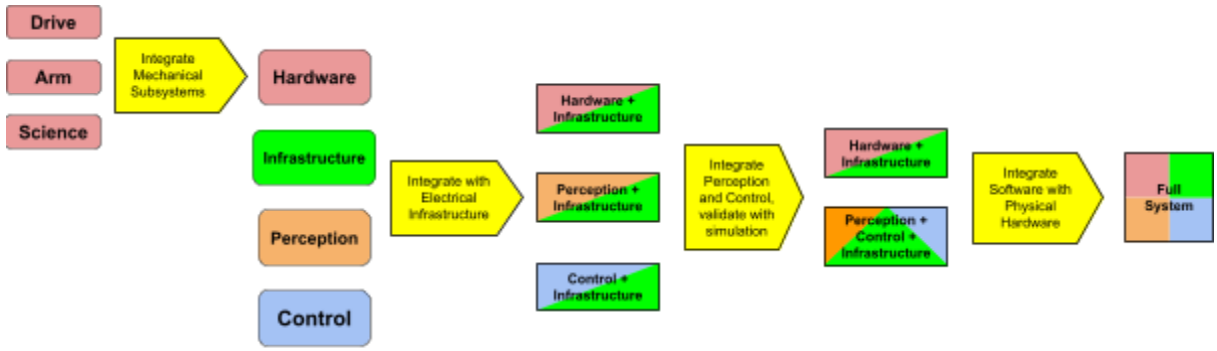

Figure 5: Integration sequencing, from individual subsystem development to full system

Status of Integration (as of 2/27)

Hardware integration: The arm and science systems are designed for modular integration with the drive system to allow rapid add/remove between events. The science system materials have been gathered, functionality confirmed, and will be deployed on the rover along with the arm.

Infrastructure integration**:** Drive and science hardware have been integrated with power distribution, and communications systems have been integrated with power distribution. Compute systems partly integrated with rover power distribution; cameras and LiDAR are not yet integrated with rover power distribution and are currently used for off-rover testing and software development.

Perception integration: A multi-camera (4-camera) pipeline streams to ROS 2 topics for use across the codebase and supports streaming to a web-based control dashboard via WebRTC/streaming. Functionality is confirmed, and the dashboard needs to be integrated with radio control, communication, and streaming onboard the rover. The Livox Mid-360 LiDAR is functioning on an Ubuntu laptop; integration into the ROS 2 rover codebase for navigation is pending.

Control integration: Navigation and manipulation are in progress; perception modules include YOLO-based object detection and SLAM-based navigation development. Manipulation remains teleop-first with planned MoveIt 2 integration, and deployment to the Jetson is planned as onboard infrastructure is finalized. Both the rover body and arm are functional with teleoperation using an Xbox controller in Genesis, with navigation and manipulation currently being tested.

Test Plan

We will conduct subsystem, integrated system, and operational tests to assess rover robustness and team readiness, iterating to improve reliability and performance for URC. Communications testing will validate command/telemetry link reliability over long distances, around large obstacles, and in non-line-of-sight conditions while recording dropouts, latency, and recovery behavior during rover operation. Mechanical integrity testing will drive the rover across diverse terrain and obstacles to evaluate suspension and wheel traction and to assess tolerance to vibration, bumps, inclines, and minor impacts, including checks for drivetrain thermal performance and fastener integrity after repeated runs. Manipulation testing will evaluate repeatability for representative servicing motions (approach, grasp, lift, place) and verify the arm can lift and transport loads exceeding 5 kg, with additional alignment checks using onboard perception for lander-style interactions. Science testing will run repeated

drill-and-eject cycles to confirm selective collection from depths greater than 10 cm and consistent transfer to the onboard science module, while verifying containment of samples/chemicals and clean-up procedures between runs. Control accuracy testing will verify teleoperation command response and stability for both driving and arm motion, including stop and recovery behavior, and will validate navigation behavior (costmaps, obstacle avoidance, and waypoint execution) as perception and localization components are integrated. Finally, operational readiness will be assessed through timed mock mission runs with enforced time limits, checklists, and defined operator roles, including rehearsals of failure scenarios (e.g., comms drops, sensor loss, actuator faults) to evaluate recovery procedures and operator consistency.

Team Organization and Outreach

From January through May, our team is organized around subsystem ownership: each technical lead is responsible for one rover subsystem and is accountable for delivering a competition-ready, integrated subsystem by the end of the academic year. These subsystems map directly onto our system architecture (see Fig. 1), spanning hardware (Drive, Arm, Science), infrastructure (Power, Communications, Compute), perception (Vision, Pose Estimation), and control (Navigation, Manipulation).

Project execution is managed through a combination of project management software (Notion), recurring weekly lead meetings, and integration milestones. Our workflow progresses from subsystem bring-up and bench validation, to Genesis-based simulation and interface verification, and then to integrated field tests and mission rehearsals. Throughout this process, leads coordinate cross-subsystem interfaces (including mechanical mounting and serviceability, electrical power distribution and communications links, and ROS 2 topic/service conventions) and maintain up-to-date documentation and test status so that integration proceeds incrementally rather than as a single, frantic end-of-semester merge.

Because we are a new team, training is integrated into the build itself across mechanical, electrical, and software workstreams. Members ramp up by contributing to scoped onboarding tasks within a subsystem, then transition into supervised integration and testing as the rover matures. Once full rover assembly is complete, we will shift emphasis toward operator training through repeated mock mission runs with defined roles and checklists, focusing on reliability, recovery procedures, and consistent operator workflows. We will also practice worst-case scenarios (e.g., communications drops, sensor failures, actuator faults) to ensure the team can diagnose issues quickly and complete tasks under competition constraints.

Outreach is supported through multiple channels. The Rice Robotics Club mentoring team supports local high-school robotics teams (remote and in-person where feasible). We also create tutorials and onboarding resources for new members and for students pursuing self-guided robotics projects, and we collaborate with other Rice engineering organizations (e.g., Rice ML and Rice AR/VR) on technical workshops and cross-team events. Externally, we present the rover at campus showcases (posters, design showcases, club fairs) and participate in broader community events to increase visibility. Internally, we run regular socials (movie/game nights, team dinners) to strengthen cohesion, maintain morale, and foster an environment where members can sustainably contribute over the full development cycle.

Task	Subsystem	Start Date	End Date	Completion	Task Owner	Project Phase
Research Rover Drive Designs	Drive	9/1/2025	9/15/2025		Anika Gupta	Research
Build 3D Printed Proof of Concept	Drive	9/15/2025	9/29/2025		Anika Gupta	Design
Design Chassis	Drive	9/29/2025	10/13/2025		Anika Gupta	Design
Prototype Wheel Designs	Drive	10/13/2025	10/20/2025		Anika Gupta	Design
Select Motors, Bearings, Suspension Components	Drive	10/20/2025	10/27/2025		Anika Gupta	Design
Complete Rover CAD Assembly	Drive	10/27/2025	11/10/2025		Anika Gupta	Design
Manufacture Rover Chassis and Wheels	Drive	11/10/2025	12/8/2025		Anika Gupta	Building/Integration
Assemble Medium-Fidelity Rover	Drive	1/19/2026	2/2/2026		Anika Gupta	Building/Integration
Develop Control Script for Drivetrain + Drill	Drive	1/19/2026	2/2/2026		Hannah Guffrey	Building/Integration
Develop and Validate Drivetrain Control Board	Drive	1/19/2026	2/2/2026		Hannah Guffrey	Building/Integration
Select High-Fidelity Components	Drive	2/2/2026	2/9/2026		Anika Gupta	Building/Integration
Test Jetson + ESP32 Handshake and Control for Drivetrain	Drive	2/2/2026	2/22/2026		Aden Briano	Testing
Finalize High-Fidelity Rover Assembly	Drive	2/9/2026	2/23/2026		Anika Gupta	Building/Integration
Assemble High-Fidelity Rover	Drive	2/23/2026	3/9/2026		Anika Gupta	Building/Integration
Test/Iterate for Driving Over Various Terrain and Obstacles	Drive	3/9/2026	4/1/2026			Testing
Research Manipulator Designs	Arm	9/1/2025	9/15/2025		Kory Sila	Research
Design Arm Mechanism	Arm	9/15/2025	9/29/2025		Kory Sila	Design
Select Motors, Links, and Arm Components	Arm	9/29/2025	10/13/2025		Kory Sila	Design
Design Linear Slide and Arm Mount	Arm	10/13/2025	10/27/2025		Kory Sila	Design
Prototype End Effector Designs	Arm	10/27/2025	11/3/2025		Kory Sila	Design
Complete Arm CAD Assembly	Arm	11/3/2025	11/17/2025		Kory Sila	Design
Prepare Manufacturing Plan	Arm	11/17/2025	12/1/2025		Kory Sila	Design
Finalize Arm CAD for Simulation	Arm	1/19/2026	1/26/2026		Kory Sila	Building/Integration
Fabricate Arm Components	Arm	1/19/2026	2/9/2026		Kory Sila	Building/Integration
Develop Control Script for Arm	Arm	2/2/2026	2/25/2026		Sean Drzewiczewski	Testing
Assemble Arm	Arm	2/9/2026	3/2/2026		Kory Sila	Building/Integration
Integrate Arm with Rover	Arm	3/2/2026	3/16/2026		Kory Sila	Building/Integration
Test/Iterate using Tele-Op for Lifting Objects and Servicing Equipment	Arm	3/2/2026	4/1/2026		Kory Sila	Testing
Research Sample Analysis Methods	Science	9/1/2025	9/15/2025		Evelyn Hardy	Research
Prototype and Test Drill Mechanism	Science	9/15/2025	9/29/2025		Rodolfo Garcia	Design
Design Emulsion Test	Science	9/29/2025	10/13/2025		Evelyn Hardy	Design
Design Linear Slide and Drill Mount	Science	10/13/2025	10/27/2025		Rodolfo Garcia	Design
Design Science Analysis Module	Science	10/27/2025	11/10/2025		Rodolfo Garcia	Design
Complete CAD Assembly for Science System	Science	11/10/2025	12/8/2025		Rodolfo Garcia	Design
Fabricate Drill Mechanism	Science	1/19/2026	2/2/2026		Rodolfo Garcia	Building/Integration
Fabricate Analysis Module	Science	2/2/2026	2/16/2026		Evelyn Hardy	Building/Integration
Develop Stepper Elevator + Science Control PCB	Science	2/2/2026	2/16/2026		Aayan Ilyas	Building/Integration
Integrate Science System with Rover	Science	2/16/2026	3/2/2026		Rodolfo Garcia	Building/Integration
Test/Iterate for Reliability of Science Analysis and Results	Science	3/2/2026	4/1/2026		Rodolfo Garcia	Testing
Research Power Distribution Systems	Power	9/1/2025	9/22/2025		Marc Alaniz	Research
Determine Power Needs of Rover Systems	Power	9/22/2025	10/6/2025		Marc Alaniz	Research
Select Batteries and Components	Power	10/13/2025	11/3/2026		Marc Alaniz	Design
Design Low-Fidelity Power Distribution	Power	11/3/2026	11/24/2025		Marc Alaniz	Design
Complete Parts List for High Fidelity Power Distribution	Power	11/24/2025	12/8/2025		Marc Alaniz	Design
Complete 48V Distribution System	Power	1/19/2026	1/26/2026		Marc Alaniz	Building/Integration
Complete <24V Distribution System	Power	1/26/2026	2/2/2026		Marc Alaniz	Building/Integration
Create PCBs to condense power systems	Power	2/27/2026	3/27/2026		Marc Alaniz	Building/Integration
Perform endurance testing to mimic competition environment	Power	2/27/2026	3/27/2026		Marc Alaniz	Testing
Revise Power Architecture based on endurance testing	Power	3/27/2026	4/27/2026		Marc Alaniz	Building/Integration
Research Radio Options and Bandwidth Restrictions	Comms	9/1/2025	10/6/2025		Sumodha Pokhrel	Research
Select 2.4GHz and 900MHz Radios and Antennas	Comms	9/22/2025	10/6/2025		Aden Briano	Design
Design Network Architecture	Comms	10/6/2025	10/27/2025		Sumodha Pokhrel	Design
Verify Prototype Communication Protocols	Comms	10/27/2025	11/24/2025		Aden Briano	Building/Integration
Select GNSS Module	Comms	1/1/2026	1/10/2026		Aden Briano	Design
Conduct 900MHz NLOS Distance Testing	Comms	1/19/2026	1/26/2026		Aden Briano	Testing
Develop RFD/GNSS Housing Board	Comms	1/26/2026	2/9/2026		Aden Briano	Building/Integration
Verify Arm Motor Control over 900 MHz	Comms	2/9/2026	2/16/2026		Aden Briano	Testing
Verify Drivetrain Control over 900 Mhz	Comms	2/9/2026	2/25/2026		Aden Briano	Testing
Verify Drill and Stepper Elevator Control over 900MHz	Comms	2/9/2026	2/25/2026		Aden Briano	Testing
Conduct GPS Testing	Comms	2/16/2026	3/2/2026		Aden Briano	Testing
Validate Science Control and TLM over 900MHz	Comms	3/2/2026	3/9/2026		Evelyn Hardy	Testing
Develop 2.4 GHz for Video Streaming	Comms	3/9/2026	3/16/2026		Sumodha Pokhrel	Building/Integration
Integrate Video Streaming with ROS2	Comms	3/16/2026	3/23/2026		Sumodha Pokhrel	Building/Integration
Integrate Arm + Drivetrain + Science Control Scripts	Comms	3/23/2026	4/1/2026		Aden Briano	Building/Integration
Develop Encoding Scheme for Controller Input to Packet Generation	Comms	3/23/2026	4/1/2026		Aden Briano	Building/Integration

Task	Subsystem	Start Date	End Date	Completion	Task Owner	Project Phase
Research Vision System Designs	Vision	9/1/2025	9/22/2025		Israel Cantu	Research
Determine Camera Layout	Vision	9/22/2025	10/13/2025		Israel Cantu	Design
Create Camera Publishing, Realsense, and LiDAR Nodes	Vision	9/22/2025	11/3/2025		Israel Cantu	Building/Integration
Implement ArUco Tag, Keyboard, and Object Detection	Vision	11/3/2025	11/24/2025		Israel Cantu	Building/Integration
Test Functionality of Cameras	Vision	11/24/2025	12/8/2025		Israel Cantu	Testing
Complete Camera Streaming System	Vision	1/19/2026	2/2/2026		Israel Cantu	Building/Integration
Implement Keyboard Pose Estimation	Vision	2/2/2026	2/16/2026		Israel Cantu	Building/Integration
Integrate LiDar with ROS2 Codebase	Vision	2/16/2026	3/2/2026		Israel Cantu	Building/Integration
Implement YOLOv8 Object Detection	Vision	3/2/2026	3/16/2026		Israel Cantu	Building/Integration
Optimize OpenCV functionality for Jetson	Vision	3/2/2026	3/16/2026		Israel Cantu	Building/Integration
Transition to D455 Cameras	Vision	3/8/2026	3/16/2026		Israel Cantu	Building/Integration
Integrate keyboard pose estimation with Arm	Vision	3/8/2026	3/16/2026		Israel Cantu	Building/Integration
Research into Isaac ROS for Vision ML Tasks	Vision	3/16/2026	3/24/2026		Israel Cantu	Building/Integration
Transition YOLOv8 to SSD-MobileNet or YOLOv8n for Jetson deployment	Vision	3/24/2026	4/1/2026		Israel Cantu	Building/Integration
Research Relevant Simulations	Simulation	9/1/2025	9/22/2025		Cameron Huang	Research
Design Low Fidelity Rover URDF and Environment	Simulation	9/15/2025	10/13/2025		Cameron Huang	Design
Test Nav2 Navigation in Genesis	Simulation	10/13/2025	11/17/2025		Cameron Huang	Testing
Integrate Vision Models with Simulation	Simulation	11/17/2025	12/8/2025		Cameron Huang	Building/Integration
Simulate Mars Environment, Terrain, Obstacles	Simulation	1/19/2026	2/2/2026		Cameron Huang	Design
Integrate Vision Tasks in Genesis	Simulation	2/2/2026	2/16/2026		Cameron Huang	Building/Integration
Integrate Teleop Movement Operations to Genesis	Simulation	2/16/2026	3/1/2026		Cameron Huang	Building/Integration
Add Arm Controls to Simulation	Simulation	3/2/2026	3/14/2026		Cameron Huang	Building/Integration
Integrate Autonmous Navigation to Simulation	Simulation	3/15/2026	4/1/2026		Cameron Huang	Building/Integration
Research Navigation Systems	Navigation	9/1/2025	9/15/2025		Rocky Ren	Research
Research Nav2 Implementation	Navigation	9/15/2025	9/29/2025		Rocky Ren	Research
Design Communication Protocol and Error Detection	Navigation	9/29/2025	10/13/2025		Rocky Ren	Design
Implement Nav2 Simulation and Cost Map	Navigation	10/6/2025	11/10/2025		Rocky Ren	Design
Demo SLAM in Turtlebot Simulation	Navigation	11/10/2025	12/8/2025		Rocky Ren	Building/Integration
Demo Telecommunication with Modem	Navigation	1/19/2026	2/2/2026		Rocky Ren	Building/Integration
Demo Navigation in Simulation Environment	Navigation	2/2/2026	2/16/2026		Rocky Ren	Building/Integration
Integrate LiDar, GPS, and Camera Systems	Navigation	2/16/2026	3/2/2026		Rocky Ren	Building/Integration
Fine Tune Navigation Algorithm	Navigation	3/2/2026	4/1/2026		Rocky Ren	Testing
Research Manpulation Control Systems	Manipulation	9/1/2025	9/22/2025		Eric Grun	Research
Research MoveIt2 Tools and Implementation	Manipulation	9/22/2025	10/6/2025		Eric Grun	Research
Implement MoveIt2 with Generic Arm URDF	Manipulation	10/6/2025	10/27/2025		Eric Grun	Design
Research Motion Planning Implementation	Manipulation	10/27/2025	11/17/2025		Eric Grun	Research
Implement Planning for Equipment Servicing Tasks	Manipulation	11/17/2025	12/8/2025		Eric Grun	Design
Set Up Server to Send Manipulation Commands to Arm	Manipulation	1/19/2026	2/2/2026		Eric Grun	Building/Integration
Incorporate URDF of Actual Arm	Manipulation	2/2/2026	2/9/2026		Eric Grun	Building/Integration
Implement Arm Position Streaming	Manipulation	2/9/2026	2/16/2026		Eric Grun	Building/Integration
Create Nodes for Teleop Arm Control and Position Streaming	Manipulation	2/16/2026	3/2/2026		Eric Grun	Building/Integration
Integrate Real Arm w/ MoveIt2 autonomous control	Manipulation	3/2/2026	3/16/2026		Eric Grun	Building/Integration
Complete Deployment to Rover Computer	Manipulation	3/16/2026	3/23/2026		Eric Grun	Building/Integration
Remote Dashboard Control Integration for Arm	Manipulation	3/23/2026	4/1/2026		Eric Grun	Building/Integration
Integrate Vision + Navigation and Verify Path Generation	Autonomous Navigation	4/1/2026	4/20/2026		Israel Cantu	Competition Prep
Integrate Vision + Navigation + Drive and Verify Autonomous Navigation	Autonomous Navigation	4/20/2026	5/4/2026		Rocky Ren	Competition Prep
Integrate Manipulation + Perception and Verify Path Planning	Equipment Servicing	4/1/2026	4/20/2026		Israel Cantu	Competition Prep
Integrate Arm + Manipulation + Perception and Automate Equipment Servicing Tasks	Equipment Servicing	4/20/2026	5/4/2026		Eric Grun	Competition Prep
Practice Tele-Op Navigation and Manipulation in Mock Competition Conditions	Extreme Delivery	4/1/2026	5/4/2026		Anika Gupta	Competition Prep
Integrate Vision + Compute for Documenting Sampling Site	Science Mission	4/1/2026	5/4/2026		Evelyn Hardy	Competition Prep
Integrate Vision + Navigation + Science for Sample Collection/Analysis	Science Mission	4/1/2026	5/4/2026		Rodolfo Garcia	Competition Prep
Full System Testing and Mock Competition Practice	Competition Logistics	5/4/2026	5/24/2026		Full Team	Competition Prep
Travel to URC	Competition Logistics	5/25/2026	5/26/2026		Full Team	Competition Prep
Compete in URC	Competition Logistics	5/27/2026	5/30/2026		Full Team	Competition Prep

Gantt Chart

Rice Robotics Rover : URC SAR BUDGET

As of February 23, 2026

Description	Amount
INCOME	
Income Received to Date	
Carryover Cash from Prior Year (C Fund Balance)	$3,911.29
Engineering School (RESC)	$2,750.00
Mechanical Engineering Department	$1,000.00
Computer Science Department	$2,000.00
Electrical Engineering Department	$1,000.00
OEDK	$2,000.00
Ken Kennedy Institute	$1,000.00
Nico Leibert (Club President)	$500.00
Anonymous	$340.00
Total Income Received to Date	**$14,501.29**
Total Additional Income Anticipated	***$2,000.00***
TOTAL INCOME	**$16,501.29**
EXPENSES	
Purchased to Date	
Rover Chassis & Drive (extrusions, motors, hardware, bearings)	$1,278.80
Rover Arm System (ROBSTRIDE motors, auger, servo shield)	$360.44
Electrical System (power converters, wiring, motor drivers, fuses)	$643.15
Communications System (RFD900x, 2.4GHz radios, GNSS x2, antennas)	$1,044.90
Compute & Electronics (Jetson Orin Nano, ESP32s, cameras, sensors)	$1,874.64
Administrative & Science (keyboard, ethanol, URC signup fee)	$249.98
Subtotal — Expenses to Date	**$5,451.91**
Planned Future Expenses (Through Project Completion)	
Custom PCB Fabrication (power distribution and sensor boards)	*$400.00*
Additional Sensors, Connectors, Wiring, Consumables, Misc. Hardware	*$420.00*
Rover Chassis Refinements (machining, finishing)	*$200.00*
URC Rover Transportation, Team Travel (shipping, flights, housing for 6)	*$3,300.00*
Subtotal — Planned Future Expenses	***$4,325.00***
TOTAL PROJECTED EXPENSES	**$9,226.91**
PROJECT FINANCIAL SUMMARY	
Estimated Total Cost of Fielded Rover System (Must be < $24,000)	**$6,476.91**
Forecast Funding Surplus/Deficit	**$6,724.38**
URC Travel Funding Available?	**Sufficient Funds**

Our team is based in Houston, Texas (humid subtropical), which differs from the arid desert terrain at MDRS. Accordingly, local testing needs to prioritize instrument functionality, detection sensitivity, and repeatability rather than replication of the Martian environment. For field testing, we seek out local sites with exposed mineral soils and low organic content (such as excavation sites, dry sandy creek beds, and caliche exposures) to approximate the sparse-biomass conditions we expect to encounter at competition. We calibrate the emulsion assay using prepared soils with known lipid concentrations (positive controls) and sterile, lipid-free substrates (negative controls), then perform field shakedown tests at exposed mineral, low-organic sites (e.g., excavation spoil, dry sandy creek beds, caliche) to approximate sparse-biomass conditions before adapting procedures to MDRS soils.

The science module integrates a drill-based sampling system, temperature and moisture sensing, and a two-chamber emulsion test. The emulsion test uses a sealed ethanol chamber for solvent extraction and a water/optics chamber for detection, connected by a pump and inline filter.

Lipids are organic compounds insoluble in water and soluble in organic solvents and are fundamental components of biological membranes [1]. Because lipid bilayers underpin cellular compartmentalization for all known terrestrial life, lipid-like compounds are considered high-value biosignature targets, and their hydrocarbon backbones can persist over geologic timescales relative to proteins or nucleic acids [1], [2]. The emulsion assay leverages lipid amphipathicity: lipids dissolved in ethanol form a milky emulsion when introduced into water, producing increased turbidity measurable with an optical sensor [1]. This approach is well-suited to the 2026 URC Science Mission, which requires rovers to investigate in-situ soils at MDRS as a Mars mission analog and to execute a documented, repeatable scientific procedure within student-team constraints [3]. Lipid biomarkers have been applied in terrestrial Mars-analog environments (e.g., hyperarid and acidic sites such as the Atacama Desert and Rio Tinto) to assess microbial signatures under harsh geochemistry [1], [4]. While flagship missions such as Curiosity's Sample Analysis at Mars (SAM) use GC-MS–based methods for detailed organic characterization [5], an emulsion test provides a robust, low-cost screening measurement for organic presence that can be executed reliably in the URC field setting [3].

For sampling, the drill conveys regolith upward into a sealed cache, targeting material at least 10 cm below the surface to reduce surficial contamination effects. A weight sensor triggers a servo closure to seal the cache and initiate transfer of soil into the ethanol extraction chamber. After a fixed soak period (~5 min), the pump moves the ethanol-soil mixture through a filter into the water/optics chamber. The optical sensor records baseline and post-injection readings; negligible change indicates low/absent extractable lipids, while increased turbidity indicates emulsion formation consistent with lipid-like compounds [1]. After drilling, the temperature and moisture sensors at the drill tip record local environmental context for each sample to support interpretation and site-to-site comparison.

References

[1] P. L. Finkel, D. Carrizo, V. Parro, and L. Sánchez-García, "An Overview of Lipid Biomarkers in Terrestrial Extreme Environments with Relevance for Mars Exploration," *Astrobiology*, vol. 23, no. 5, pp. 563-604, May 2023.

[2] F. Westall et al., "Biosignatures on Mars: What, Where, and How? Implications for the Search for Martian Life," *Astrobiology*, vol. 15, no. 11, pp. 998-1029, Nov. 2015.

[3] The Mars Society, "University Rover Challenge 2026 - Requirements and Guidelines," 2025. [Online]. Available: http://urc.marssociety.org.

[4] C. R. Stoker, "Life on Mars, can we detect it?," *Nature Communications*, vol. 14, Art. no. 807, Feb. 2023.

[5] P. R. Mahaffy et al., "The Sample Analysis at Mars Investigation and Instrument Suite," *Space Science Reviews*, vol. 170, pp. 401-478, 2012.